FAST EXPONENTIATION

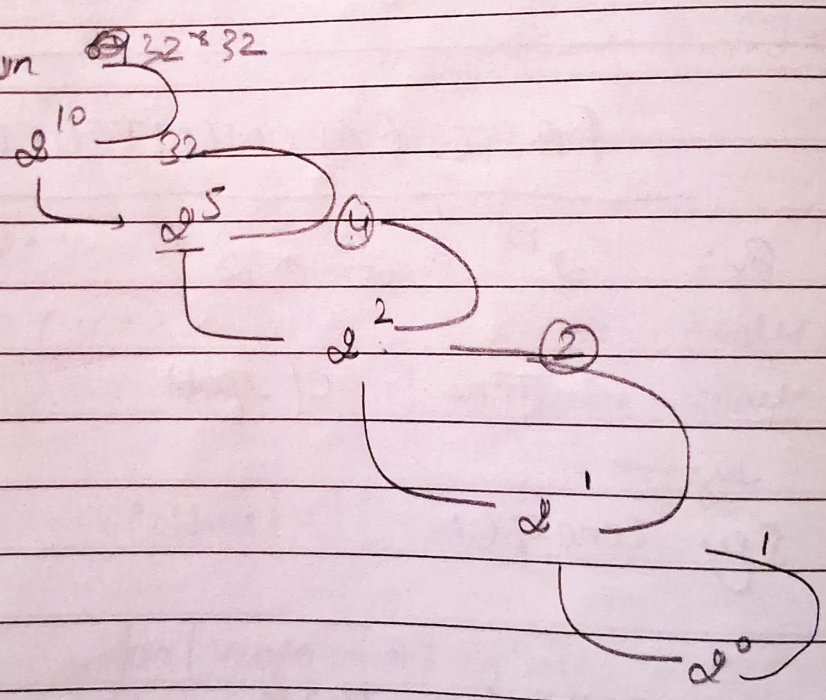
Ex: $2^{10} = 2^5 * 2^5$

Time = $O(\log b)$

Key Concept:

$$a^b = \begin{cases} (a^{b/2})^2 & \text{, if } b \text{ is even} \\ a * (a^{b/2})^2 & \text{, if } b \text{ is odd.} \end{cases}$$

Ex: 3^{12} $a = 3$, $b = 12$ (even)

```
findPower( int a, int b) {
    if (b == 0) return 1;
    int half = findPower( a, b/2)
    int result = half * half
    if ( b % 2 == 1)
        result = ( result * a)
    return result;
}
```

Dry run